

3	(a)		Resultant Force, $F_R = ma = 4000 (0.30) = 1200 \text{ N}$ Applied force, $F = F_R + f = 1200 + 700 = 1900 \text{ N}$ Therefore power = $1900 (8.0) = 15200 \text{ W}$	
	(b)	(i)	Applied force, $P = Fv$ $7600 = 700 v$ $F = f$, since max speed, $a = 0$ $V = 10.9 \text{ m s}^{-1}$	
		(ii)	$\sin \theta = 1/25$ Total opposing force $f_T = f + mg \sin \theta = 700 + 39240 (0.04) = 2270 \text{ N}$ $P = Fv$ $v = 7600 / 2270 = 3.35 \text{ m s}^{-1}$	